

Lab 1: Introduction to Maven and Gradle: Overview of Build Automation Tools, Key Differences Between Maven and Gradle, Installation and Setup.

Introduction to Maven and Gradle

Overview of Build Automation Tools

Build automation tools help developers streamline the process of building, testing, and deploying software projects. They take care of repetitive tasks like compiling code, managing dependencies, and packaging applications, which makes development more efficient and error-free.

Two popular tools in the Java ecosystem are **Maven** and **Gradle**. Both are great for managing project builds and dependencies, but they have some key differences.

Maven

- Maven is a build automation tool primarily used for Java projects. It uses an XML configuration file called pom.xml (Project Object Model) to define project settings, dependencies, and build steps.
- **Main Features:**
 - Predefined project structure and lifecycle phases.
 - Automatic dependency management through Maven Central.
 - Wide range of plugins for things like testing and deployment.
 - Supports complex projects with multiple modules.

Gradle

- Gradle is a more modern and versatile build tool that supports multiple programming languages, including Java, Groovy, and Kotlin. It uses a domain-specific language (DSL) for build scripts, written in Groovy or Kotlin.
- **Main Features:**
 - Faster builds thanks to task caching and incremental builds.
 - Flexible and customizable build scripts.
 - Works with Maven repositories for dependency management.
 - Excellent support for multi-module and cross-language projects.

- Integrates easily with CI/CD pipelines.

Key Differences between Maven and Gradle

Aspect	Maven (pom.xml)	Gradle (Groovy/Kotlin DSL)
Configuration	XML (pom.xml)	Groovy or Kotlin DSL
Performance	Slower	Faster (caching)
Flexibility	Less flexible	Highly customizable
Ease of Learning	Easier to pick up	Steeper
Script Size	Larger	Smaller
Dependency Management	Maven Central	Compatible with Maven
Plugin Support	Large ecosystem	Extensible ecosystem

Installation and Setup

How to Install Maven:

1. Download Maven:

- Go to the <https://maven.apache.org> and download the latest binary ZIP file.

2. Extract the ZIP File:

- Right-click the downloaded ZIP file and select **Extract All...** or use any extraction tool like WinRAR or 7-Zip.

3. Move the Folder:

- Create “maven_batchA” folder inside C:\Program Files\
- After extraction, move the extracted **Maven file** (usually named **apache-maven-x.x.x**) to a directory like C:\Program Files\maven_batchA

4. Navigate to the bin Folder:

- Open the **Maven folder**, then navigate to the **bin** folder inside.

- Copy the path from the File Explorer address bar(e.g., **C:\Program Files\maven_batchA\apache-maven-x.x.x\bin**).
5. **Set Environment Variables:**
 - Open the **Start Menu**, search for **Environment Variables**, and select **Edit the system environment variables**.
 - Click **Environment Variables**.
 - Under **System Variables**:
 - Find the **path**, double click on it and click **New**.
 - Paste the full path to the bin folder of your Maven directory (e.g., **C:\Program Files\maven_batchA\apache-maven-x.x.x\bin**).
 6. **Save the Changes:**
 - Click **OK** to close the windows and save your changes.
 7. **Verify the Installation:**
 - Open Command Prompt and run: **mvn -v** If Maven is correctly installed, it will display the version number.

How to install Gradle

1. **Download Gradle:**

Visit the www.gradle.org and download the latest binary ZIP file.
2. **Extract the ZIP File:**
 - Right-click the downloaded ZIP file and select **Extract All...** or use any extraction tool like WinRAR or 7-Zip.
3. **Move the Folder:**
 - After extraction, move the extracted **Gradle folder** (usually named **gradle-x.x.x**) to a convenient directory like **C:\Program Files**.
4. **Navigate to the bin Folder:**
 - Open the **Gradle folder**, then navigate to the **bin** folder inside.
 - Copy the path from the File Explorer address bar (e.g., **C:\Program Files\gradle-x.x\bin**).
5. **Set Environment Variables:**
 - Open the **Start Menu**, search for **Environment Variables**, and select **Edit the system environment variables**.
 - Click **Environment Variables**.
 - Under **System Variables**:
 - Find the **path**, double click on it and click **New**.

- Paste the full path to the bin folder of your Gradle directory (e.g., **C:\Program Files\gradle-x.x.x\bin**).

6. Save the Changes:

- Click **OK** to close the windows and save your changes.

7. Verify the Installation:

- Open a terminal or Command Prompt and run: **gradle -v** If it shows the Gradle version, the setup is complete.